APU CLOUD INNOVATION HACKATHON 2026
SMART CAMPUS TRACK

Spotter

The APU SweatZone, connected.

Clock in with a scan, log every lift, eat smart from the campus cafeteria and climb the ranks. One app for every student at the gym.



Gym life still runs on guesswork



Check-in on paper
Students queue at the counter to show an ID card.



Arriving blind
No way to know how packed the SweatZone is before walking over.



Lifts in a notebook
Routines, PRs and progress live in notes apps and memory.



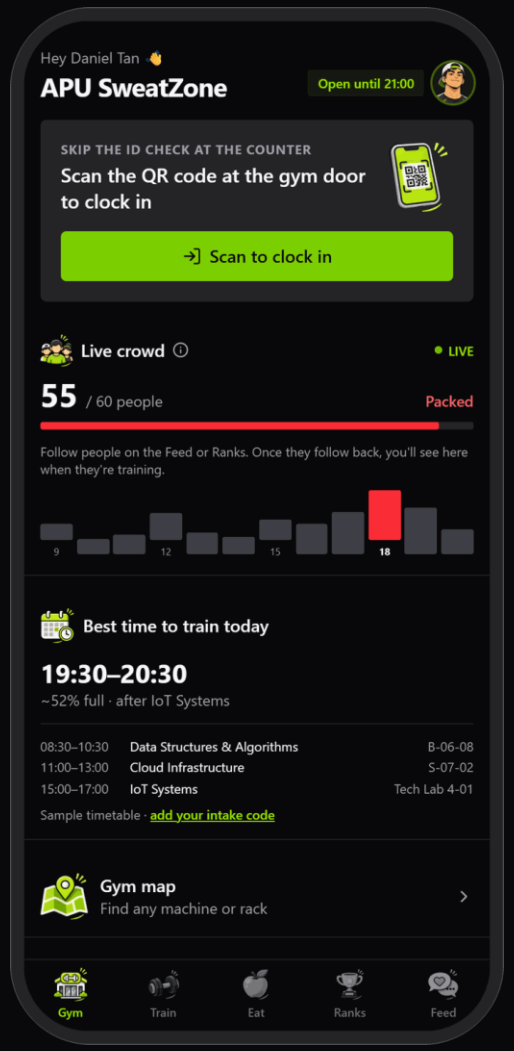
Calories by guesswork
Cafeteria dishes have no macros, so eating for a goal is a guess.

THE GAP

Nobody knows who trains, who lifts what, or who could represent APU.

The SweatZone has no shared record of its students: no head count, no lift history, no way to spot talent.

From the door to your last set



Clock in with a scan
The door QR replaces the ID check

Live crowd
Head count and the quietest hour

3D gym map
Find any machine in the SweatZone

Train
Routines, sets, rest timer and PRs

Muscles hit
A 3D body model of what you trained

Eat smart
43 APU cafeteria dishes with macros

Ranks
13 events, DOTS and weight classes

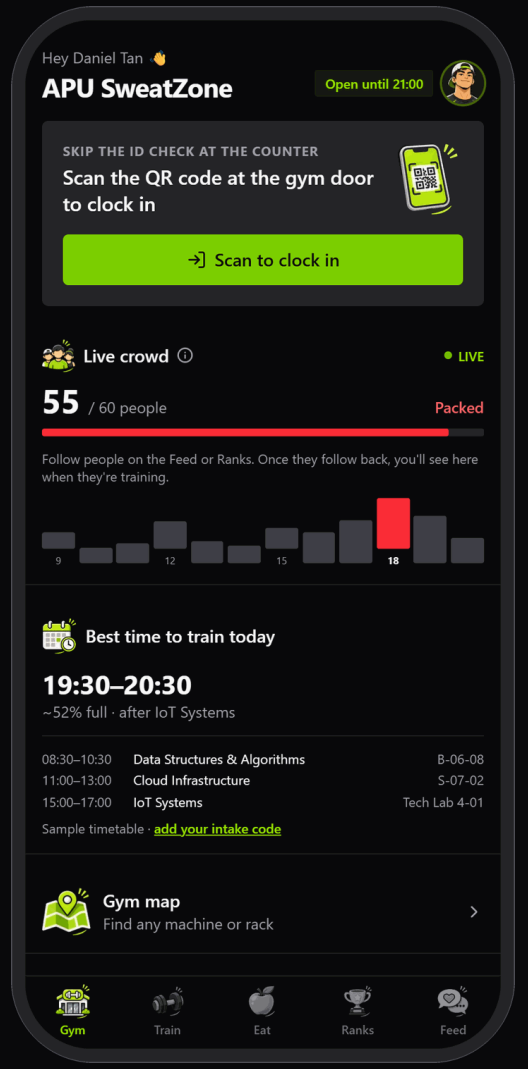
Social feed
Share, like, comment and follow

Clock in with a scan

Scan the door QR
A poster on the SweatZone door; one scan clocks you in. No more ID check at the counter.

A live head count
Every clock-in and clock-out feeds a live count out of 60 people.

Opening hours built in
Shows Open until 21:00, or when it opens, so nobody walks over to a closed gym.







Know how busy it is, and when to go

Live crowd
A running count out of 60 and an hourly chart, labelled Quiet, Busy or Packed.

Best time to train
Picks the quietest free hour between your classes, from APU's public timetable feed.

Just add your intake code
Your classes and rooms appear on the home screen next to the free slot.

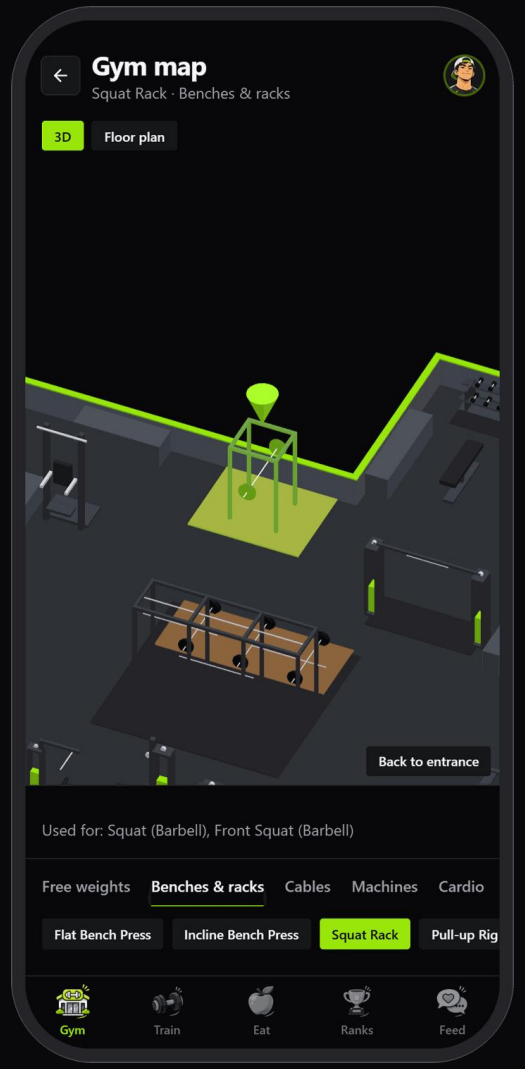
Find any machine on a 3D map

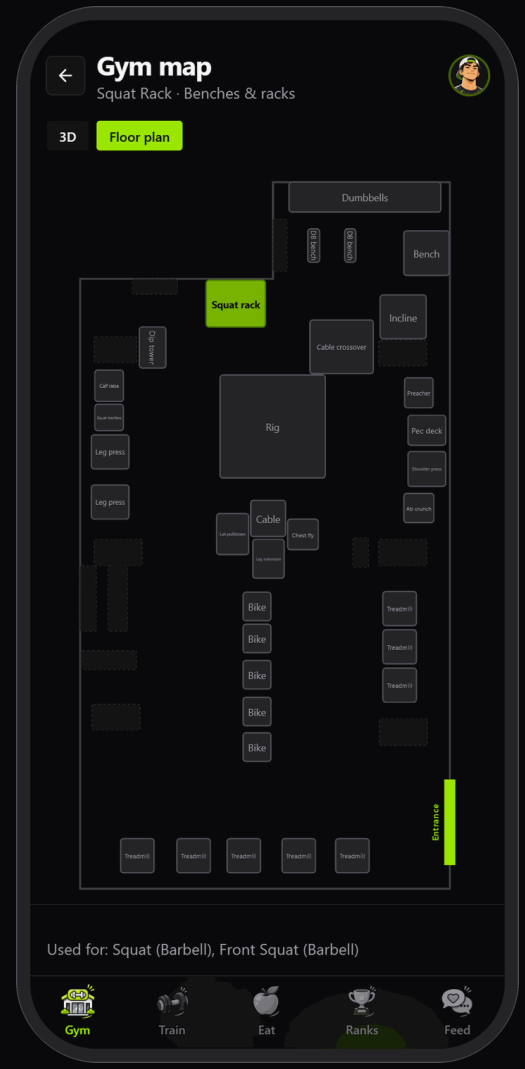
The whole SweatZone in 3D
All 34 machines, racks and benches, modelled in Blender and placed where they stand.

Tap or pick a machine
Choose by zone (free weights, racks, cables, machines, cardio) and the camera glides to it.

Exercises point to machines
Every exercise shows the machine that does it, one tap from your workout.

Floor plan view
Switch to a flat plan with the entrance marked.





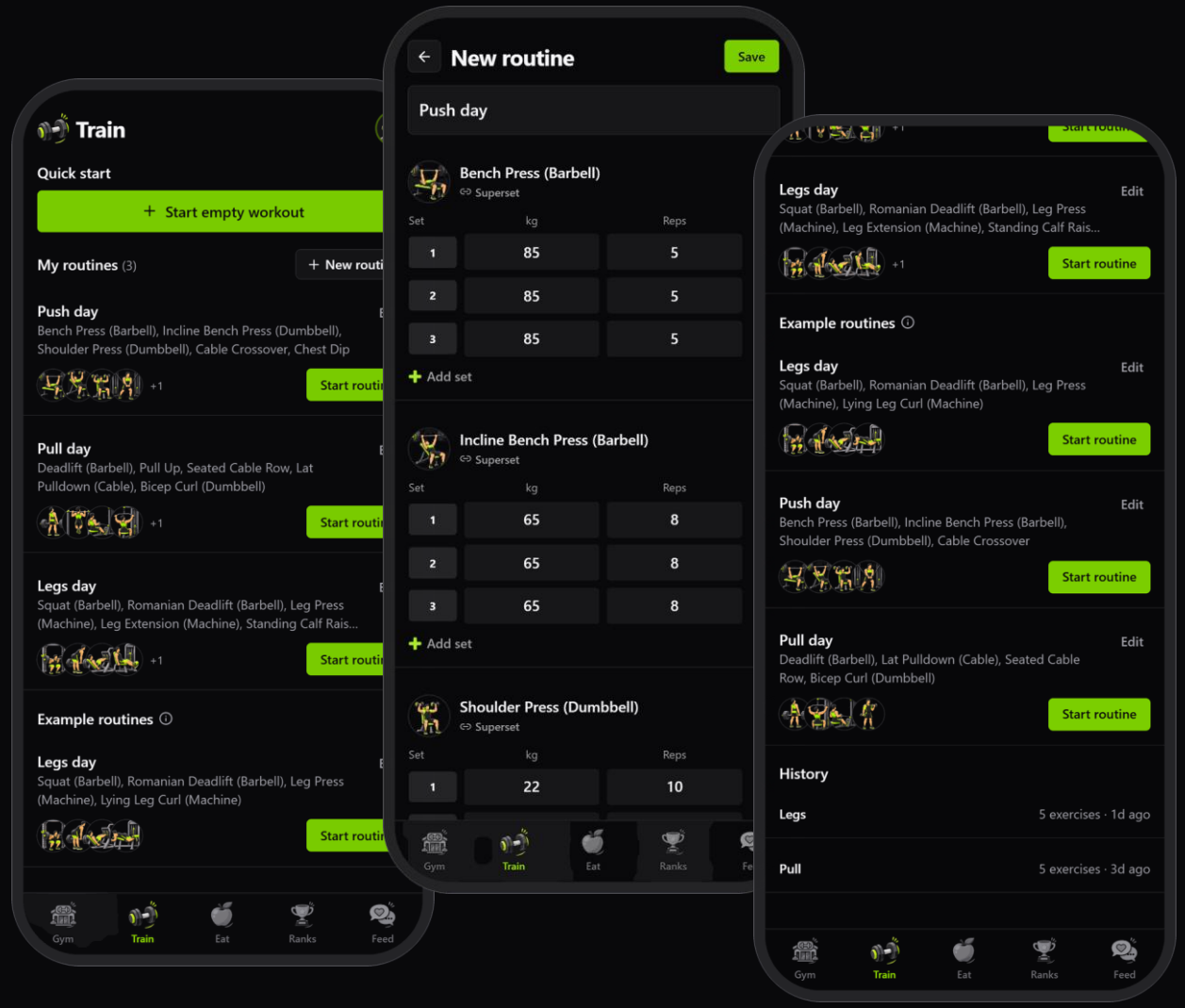
Routines that start in one tap

Quick start
Start an empty workout, or pick a routine and go.

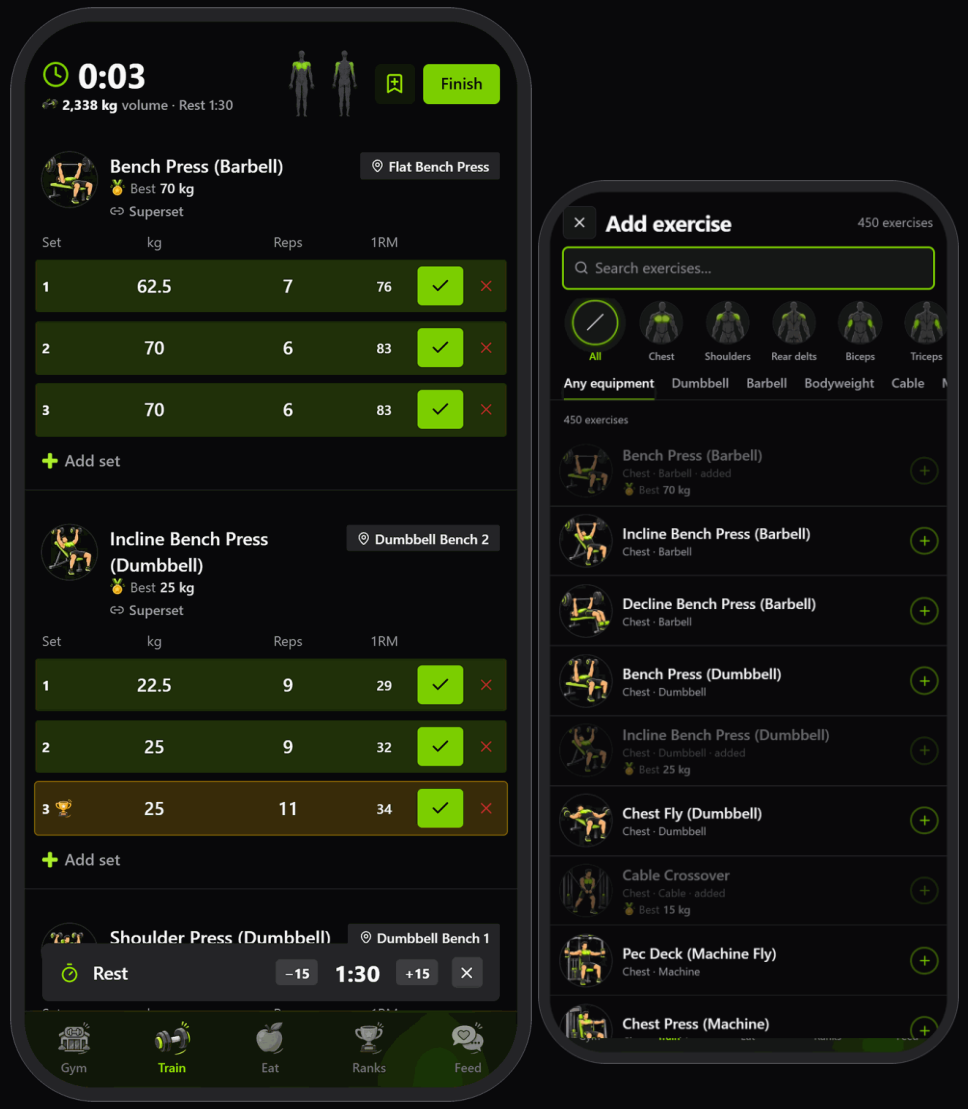
Build your own
Add exercises, sets, weights and supersets, then save the routine.

Examples to start from
Push, Pull and Legs templates you can edit and make your own.

History
Your last two sessions at a glance; the rest lives in your profile.



Log every set



Sets with kg, reps and 1RM
Tick a set to log it; each one shows an estimated one-rep max.

Rest timer and supersets
The timer starts when you tick a set (±15 s). Link exercises into supersets.

Seven set types
Warm-up, failure, drop set, rest-pause, deload, assisted and partial reps.

PR detection
Beat your history and a fanfare, a vibration and a gold badge celebrate it.

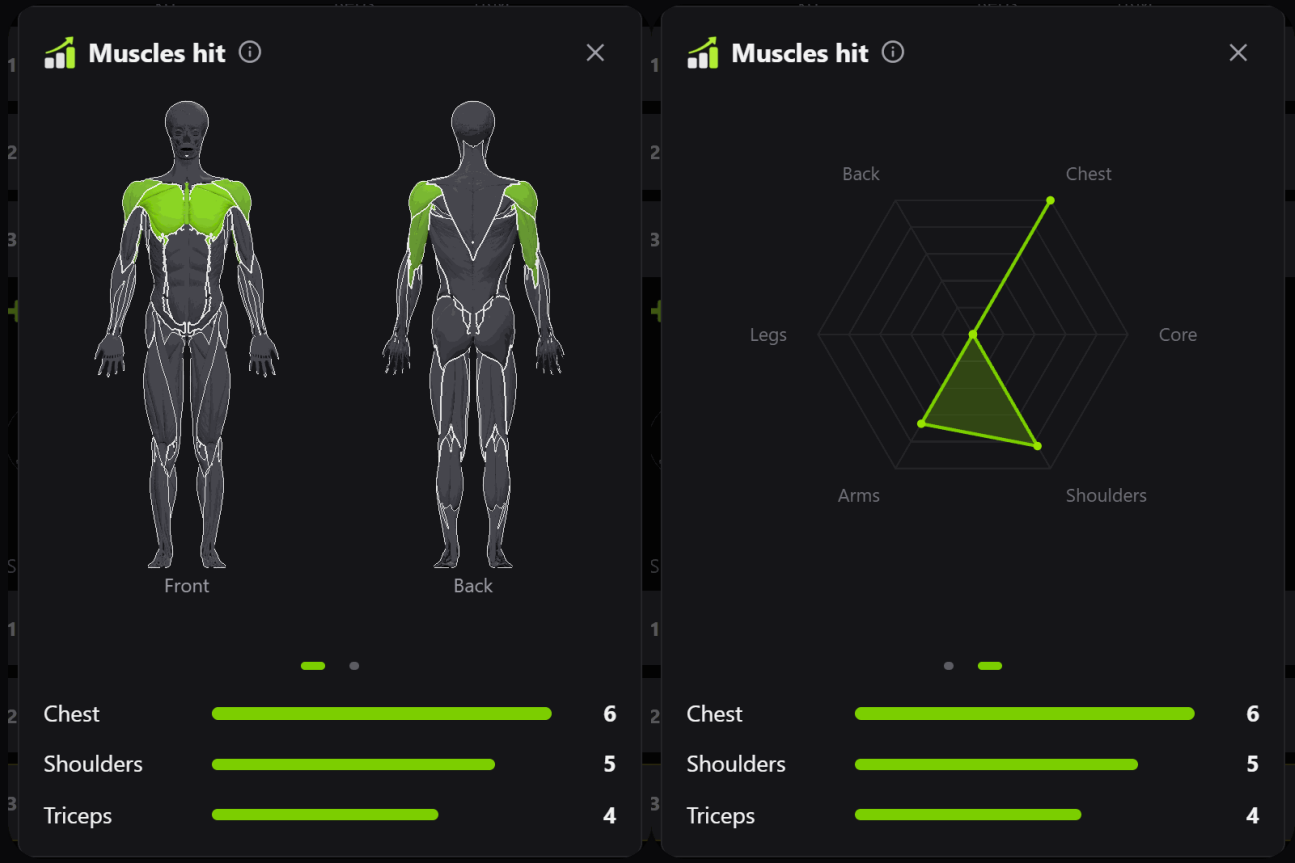
Finish and share
Add a title, photo or video, pick who sees it, and it posts to the feed.

See exactly what you trained

A 3D body model
Front and back views light up the muscles you worked, brighter for more sets.

Balance radar
Swipe to a radar chart to see which areas you are neglecting.

Trends over time
Muscle distribution over a week, a month, three months or a year in your stats.



Muscles hit from a live workout: the model, then the radar



450 exercises, built from scratch

Our own dataset
Every exercise written up and illustrated for Spotter, with its primary and secondary muscles.

Filter by muscle and equipment
Search by muscle group or equipment: barbell, dumbbell, cable, machine, bodyweight.

Progress per exercise
Charts of heaviest weight, estimated 1RM and volume, plus personal and set records.

Strength level
Compares your best 1RM with your bodyweight, from Beginner to Elite.

Eat smart from the APU cafeteria

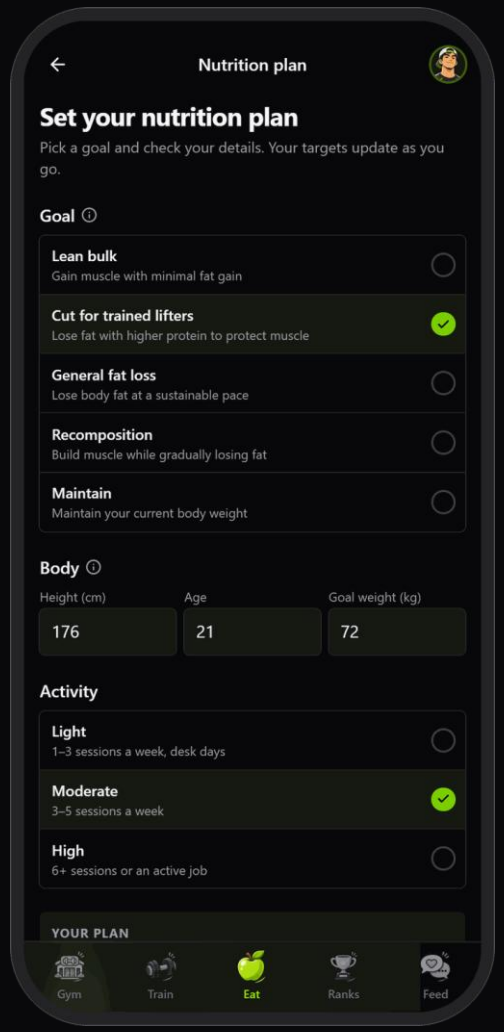
43 dishes from 9 stalls
Western, Indian, Malaysian, Arabic, Japanese, Kopitiam, Chinese and Burger, with calories and macros.

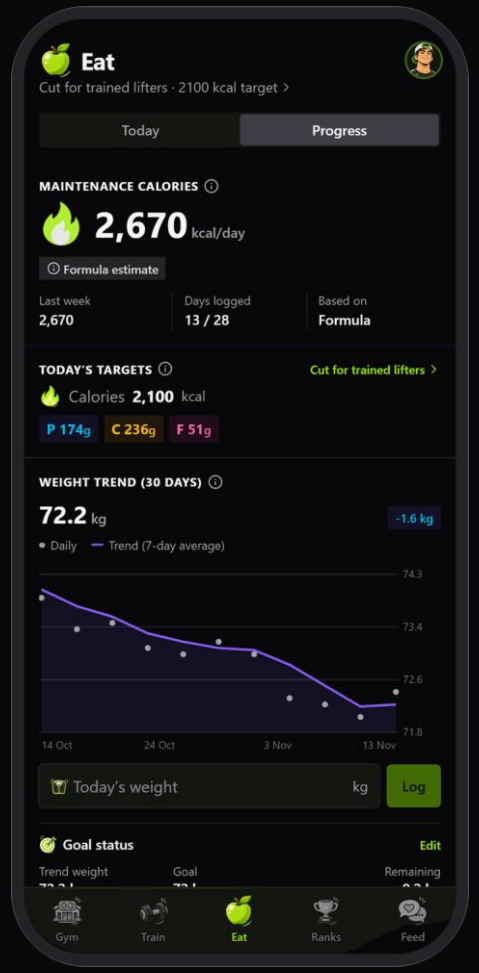
Search and filter
Find a dish by name or stall, or filter for high protein.

Snap your meal
Photograph your plate and AI (Gemini) estimates the calories and macros.

One tap to log
Add it to today and watch your calories and macros fill up.







Nutrition that follows your goal

Pick your mission
Lean bulk, cut for trained lifters, general fat loss, recomposition or maintain.

Body and activity
Height, age, weight and a light, moderate or high activity level set your daily targets.

Maintenance calories
Estimated from your own food and weight logs, with a formula as the fallback.

Weight trend and goal
A daily weight log with a seven-day trend line and a goal weight.

Climb the ranks

13 events
Powerlifting (squat, bench, deadlift, total), strength, calisthenics and athletics.

Fair for everyone
Rank open, pound for pound (DOTS) or by IPF weight class, men and women separately.

By faculty
Filter the board to a faculty and see who is strongest across campus.

How far to the next rank
A line like '150 kg to pass Kenji T.' shows exactly what it takes.



What APU gets from Spotter



Records of every lifter

Who trains, how often and what they lift, kept per student with their course, faculty and intake.



Competition selection

Rank by event, weight class and faculty to pick lifters for inter-university competitions.

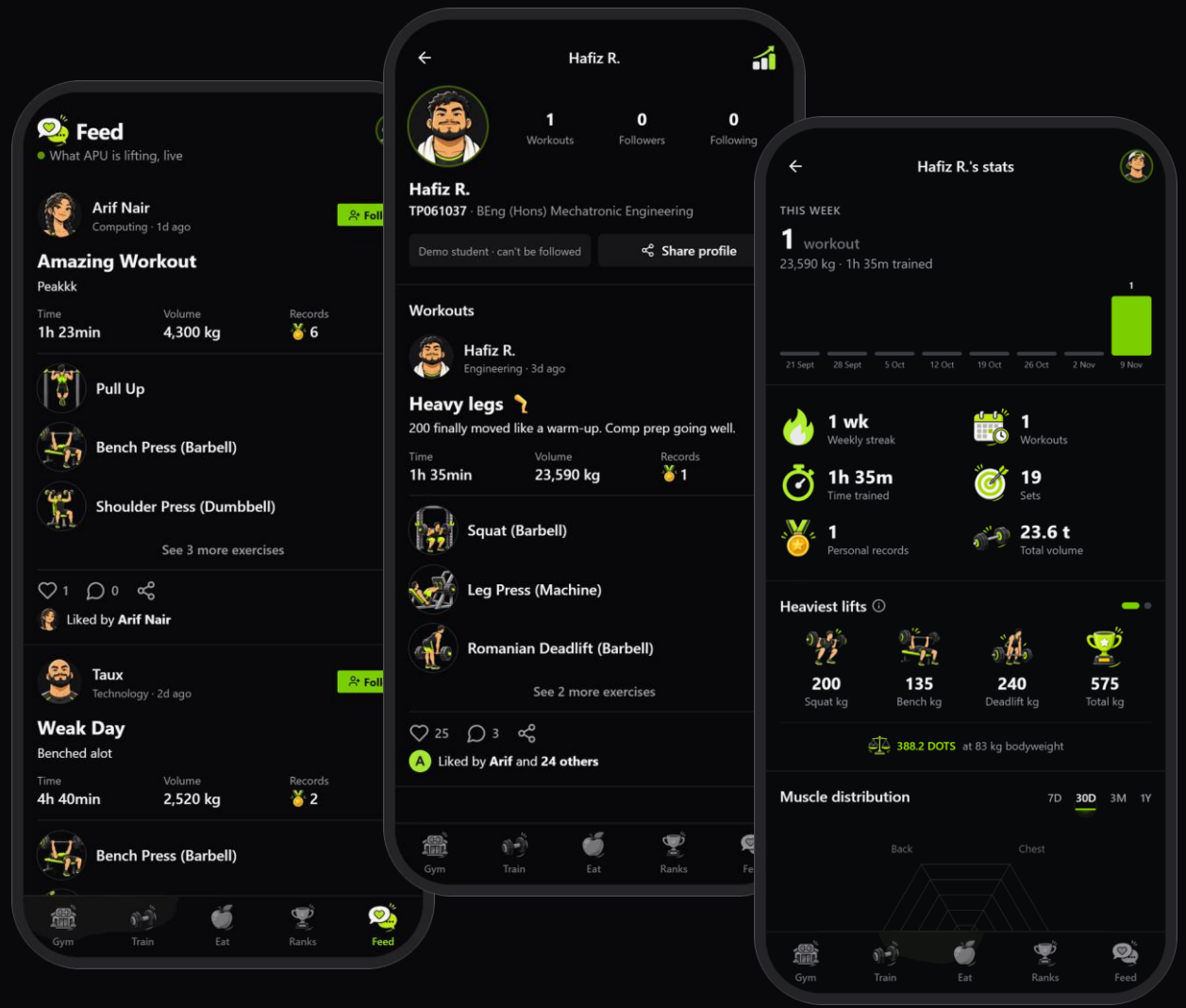


Representation and reach

Students are seen and recognised, and APU can reach and represent more of them.

Plus a live head count of the SweatZone, for planning opening hours and equipment.

A gym with a social side



A live feed

Finished workouts post with volume, time, PR count and the exercises, as they happen.

Likes and comments

React to friends' sessions; likes and comments appear in real time.

Follow and follow back

Follow classmates; once they follow back you can see when they are training.

Profiles with real stats

Workouts, weekly streak, time trained, total volume, heaviest lifts and DOTS.

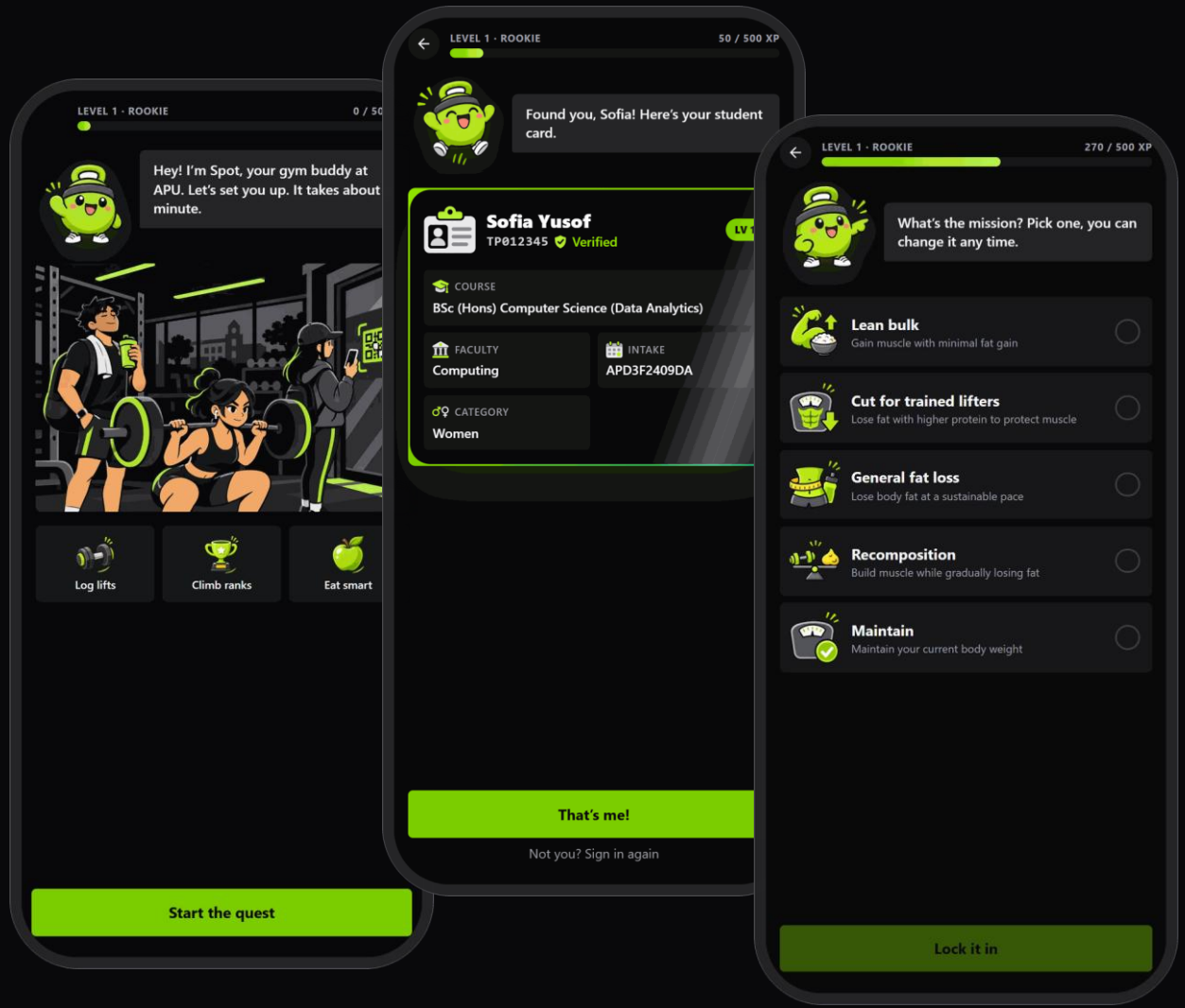
Onboarding is a quest

Spot, your gym buddy
A friendly mascot talks you through setup one step at a time, with XP for each step.

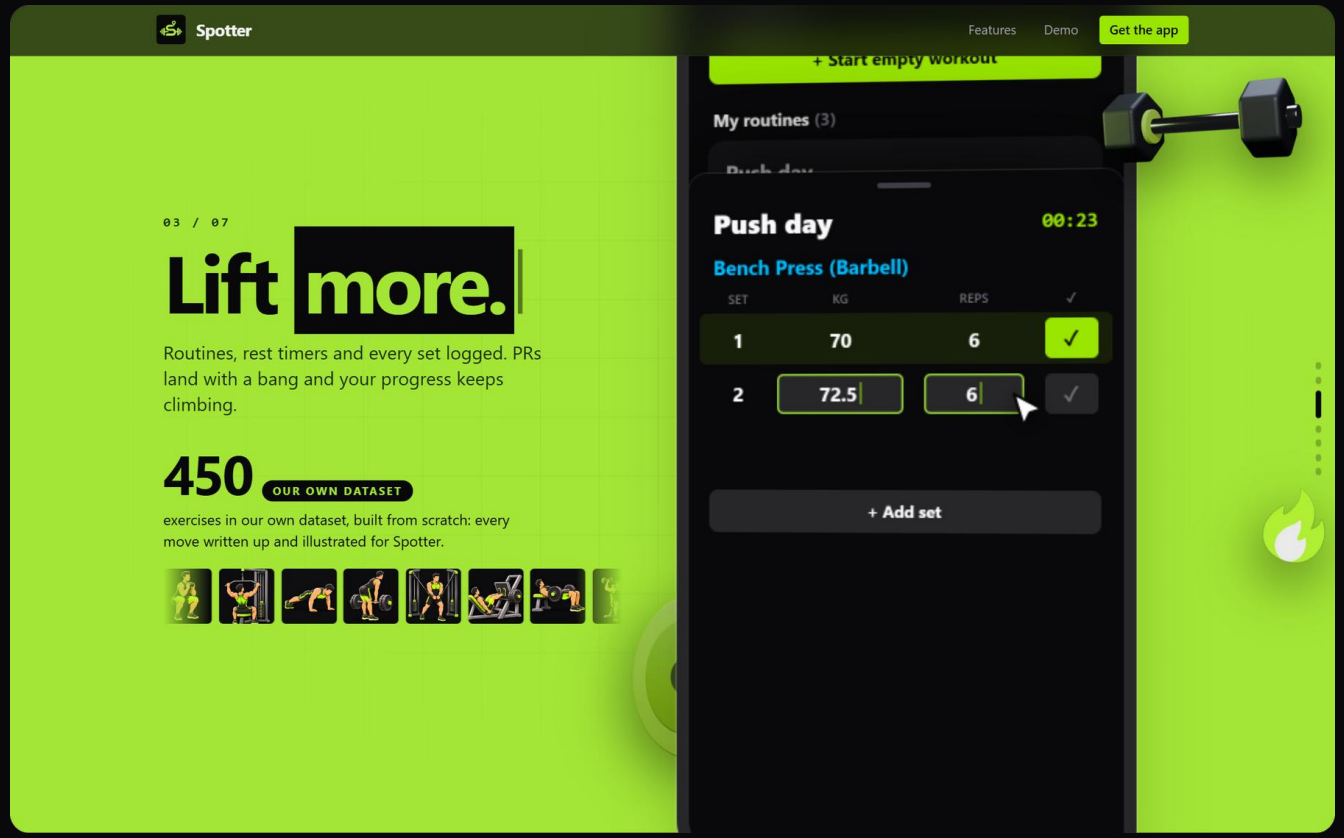
Sign in with APU
Enter your TP number and your student card appears with course, faculty and intake.

Set your mission
Body weight, PRs, goal and activity turn into a daily nutrition plan.

Change it any time
Edit your profile, units (kg or lb) and plan from Settings.



On the web and on Android



Website
A Next.js app that works in any phone browser, with a landing page and a live demo.

Android app
A native Expo app with camera QR clock-in and haptics, from the same shared logic.

Shared logic
Lifting maths, DOTS, nutrition and the 3D gym live in one TypeScript library.

Demo mode
Try the whole app on a made-up student account; nothing is saved.

How it is built

CLIENTS

Website
Next.js 15 (App Router) and Tailwind v4

Android app
Expo and React Native, with camera QR and haptics

SHARED LIBRARY

One TypeScript library
Lifting maths and PR rules
Strength standards, DOTS, IPF classes
Nutrition plans and targets
The 3D gym scene (three.js)
The onboarding quest

SERVICES

Supabase
Postgres with row-level security, anonymous auth, Realtime

Gemini
Meal photo to calories and macros

APU timetable feed
Public feed, looked up by intake code

Blender to glTF
The 3D SweatZone model, drawn with three.js

Spotter in numbers

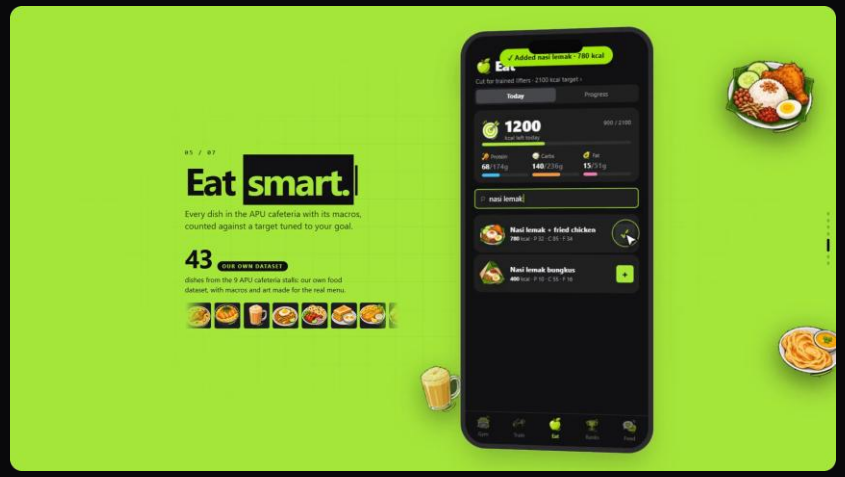
450
exercises we wrote and illustrated

43
dishes from 9 APU cafeteria stalls

34
machines placed in the 3D gym

13
events on the leaderboards

2
platforms: website and Android



Watch the 49-second demo video

Every screen in this deck is the real app, running on a demo account.

Built for the APU Cloud Innovation Hackathon 2026, Smart Campus track
github.com/tauxhd/spotter